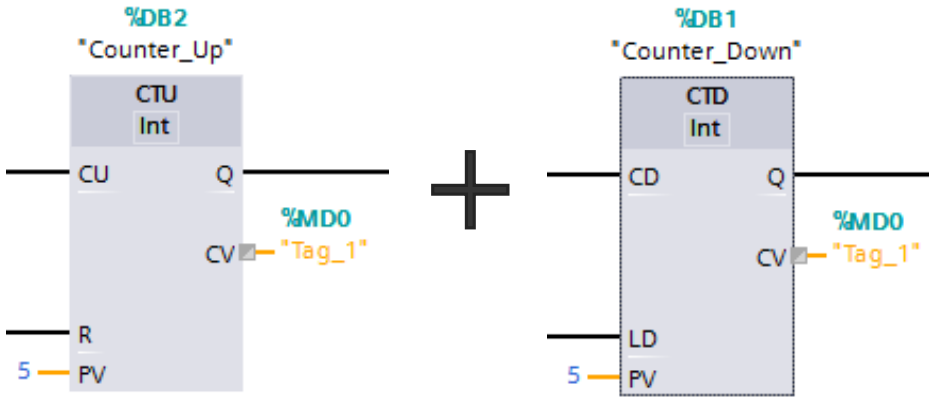
Siemens S7-1200

CPU 1212C AC/DC/Relay

Counter Operations (CTUD)

- **Count Up-Down**
- **Exercise Example**



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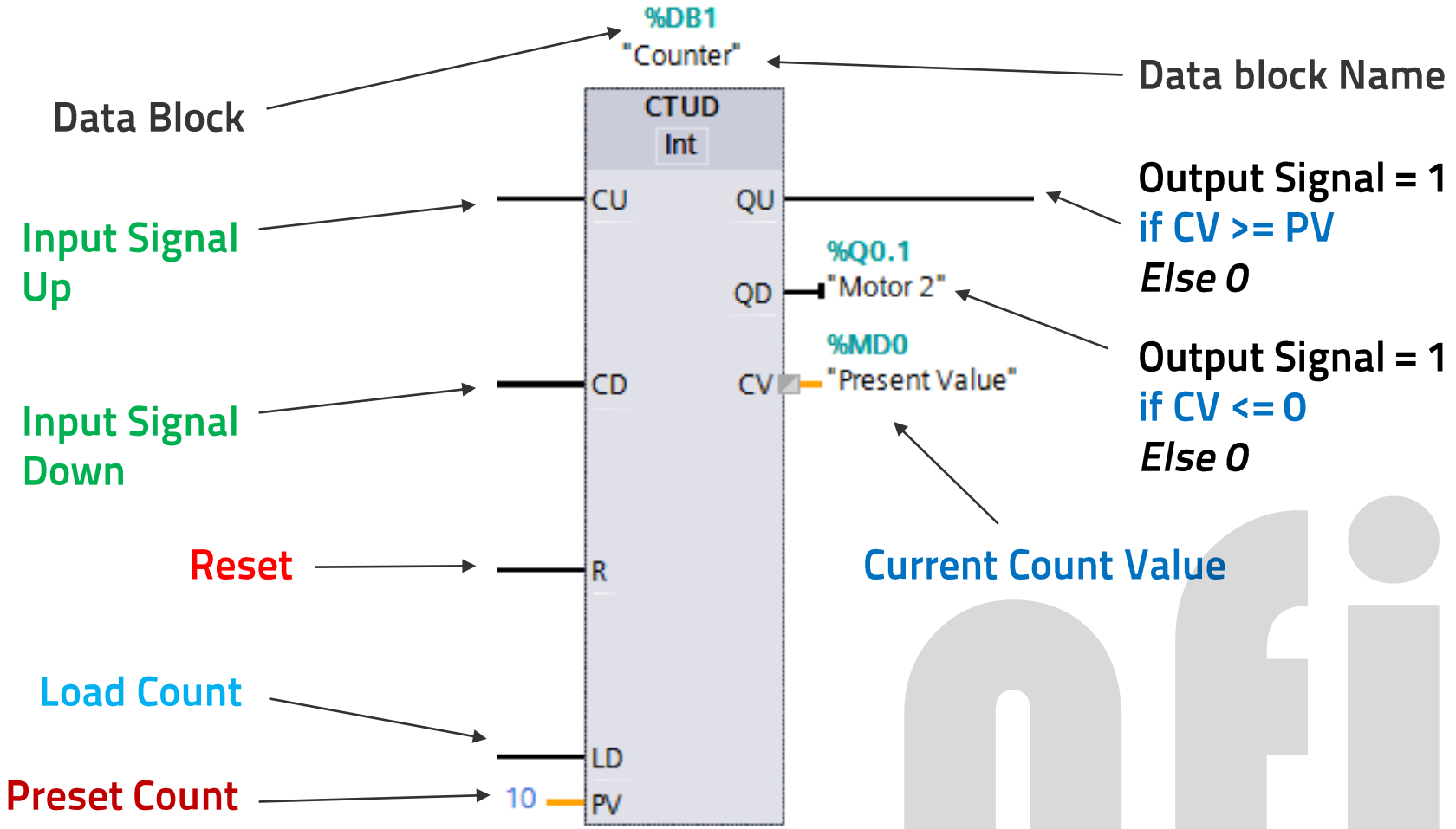


Counter Operations in LAD – **CTUD (Count Up Down)**

Count Up-Down

The "count up and down" counter (CTUD) counts up or down by 1 on the ***0 to 1 transition of the count up (CU) or count down (CD) inputs.***

Counter operations	
CTU	Count up
CTD	Count down
CTUD	Count up and down

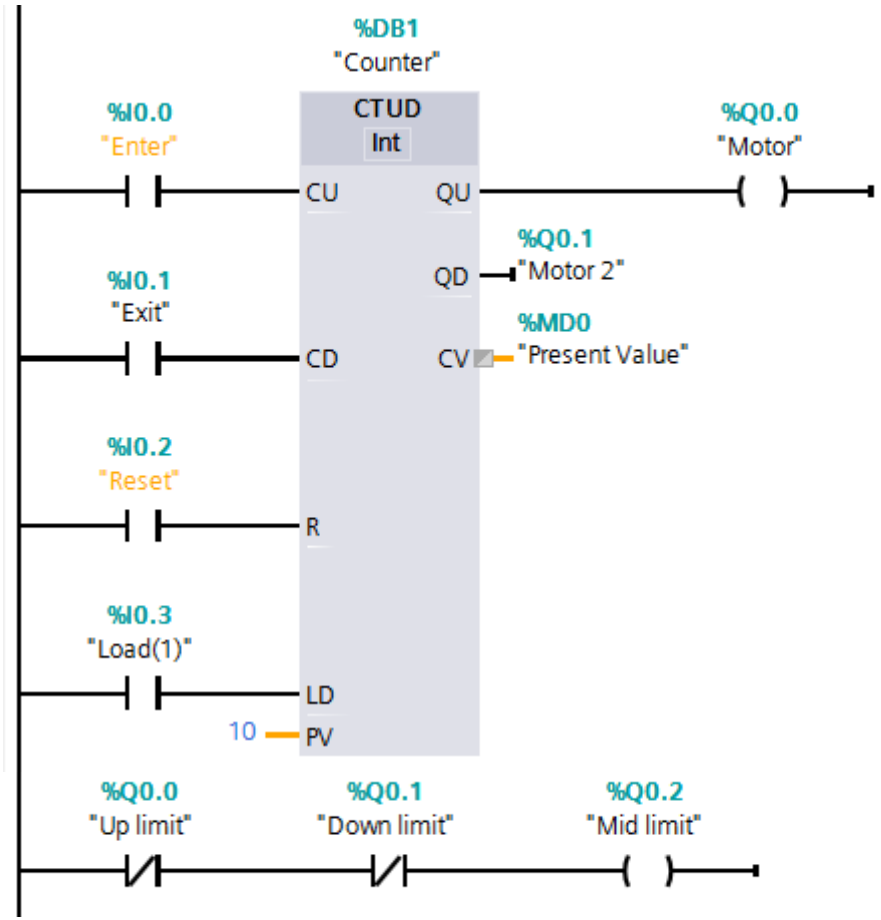




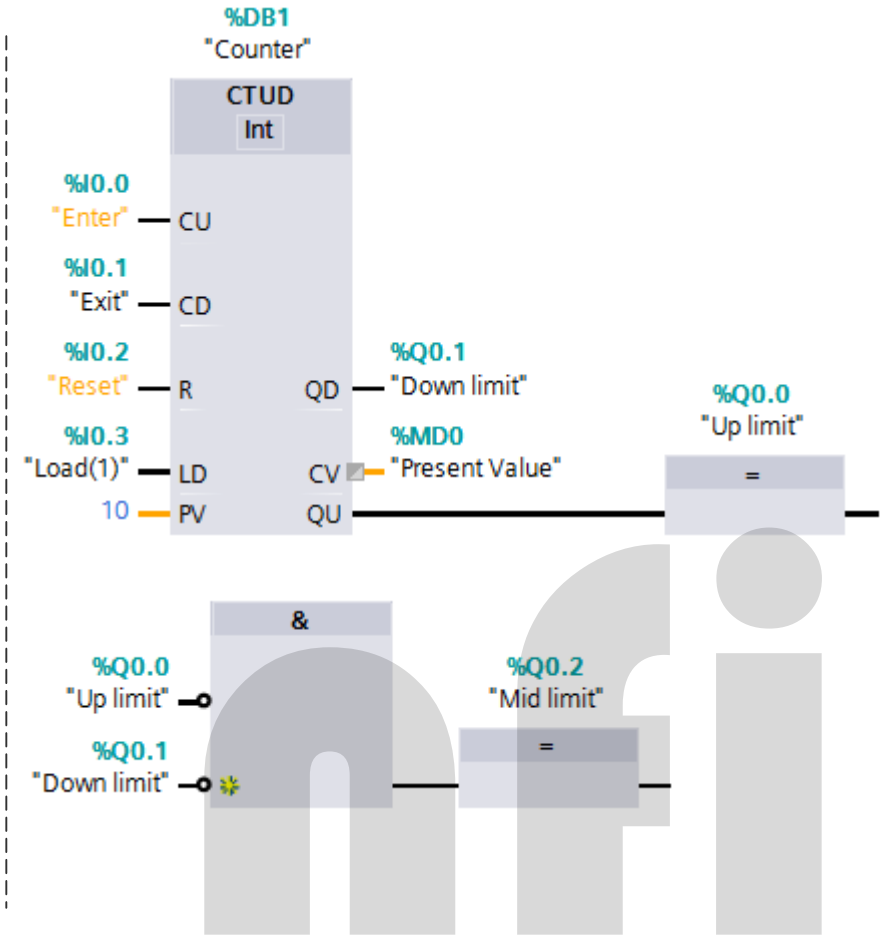
Working Example

Test the operation of Counter Up/Down using Ladder and FBD. Take PV as 10.
Take another output which should be ON if the PV is between 0 ~ 10.

Ladder Solution



FBD Solution



What did we learn in this lesson?

- If the value of **CV (current count value)** is **equal to or greater than** the value of **PV (preset value),** then the **counter output QU = 1**.
- If the value of **CV** is less than or equal to zero, then the **counter output QD = 1.**
- If the value of **LOAD changes from 0 to 1,** then the value at **PV is loaded to the counter as the new CV.**
- If the value of the reset R is changes from 0 to 1, **CV is reset to 0.**

Thank you

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